



الجامعة الألمانية الأردنية
German Jordanian University

Engineering Economics Project Proposal

School of Applied Technical Sciences

DS-301

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Executive Summary

This report presents an engineering economics proposal focused on optimizing the current lubrication approach which leads to frequent machine breakdowns, excessive oil consumption, and higher maintenance labor costs, due to time-based oil changes and a lack of proper filtration. Also face inconsistent oil quality in our machinery, resulting in premature component wear and costly part replacements. The strategy involves a cost-benefit comparison between time-based oil changes and their associated costs or applying lubrication best practices (LBP) to change the oil condition based.

Problem Statement

Oil sample analysis and component inspections revealed frequent hydraulic pump failures, driven by strict hour-based oil changes and lack of filtration. This approach leads to increased oil waste, environmental impact, and high maintenance costs. To address this, we compare two alternatives: the current time-based method vs. a condition-based strategy following Lubrication Best Practices (LBP), aiming to identify the more cost-effective and reliable solution.

Objectives

- Comparing the life-cycle cost of the 2 alternatives.
- To evaluate which strategy provides the best balance between cost efficiency and oil consumption over a multi-year horizon.
- Develop a decision-making framework for periodic maintenance based on system criticality, effect on salvage value of the components and waste oil, and cost factors.
- Reduce unscheduled downtime related to lubrication which has its own costs

Project Scope

This project will initially focus on improving hydraulic oil consumption and oil management for our main production lines in the main factory. Key LBP areas to be addressed include proper oil storage and handling, contamination control, basic in-house oil analysis, and technician training. Implement a comprehensive

solution designed to maximize component lifespan and deliver significant, sustainable cost reductions.

Methodology

Conducting a baseline assessment of current lubrication practices and consumption. Then, implement new procedures for oil storage, filtration, and dispensing. Oil control products and systems and lastly provide comprehensive training to the maintenance staff. We will take MARR 12% , and the payback value will be confirmed later according to data

Company Context

Modern Systems Co. is a leading provider of industrial equipment and motion control solutions. A similar lubrication optimization study was recently conducted for one of our clients, resulting in measurable cost savings and improved equipment reliability. This project applies the same approach internally to enhance our own operations and support long-term sustainability.